

1. NAME OF THE MEDICINAL PRODUCT

Trimbow 100 micrograms/6 micrograms/12.5 micrograms pressurised inhalation, solution

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each delivered dose (the dose leaving the mouthpiece) contains 87 micrograms of beclometasone dipropionate, 5 micrograms of formoterol fumarate dihydrate and 9 micrograms of glycopyrronium (as 11 micrograms glycopyrronium bromide).

Each metered dose (the dose leaving the valve) contains 100 micrograms of beclometasone dipropionate, 6 micrograms of formoterol fumarate dihydrate and 10 micrograms of glycopyrronium (as 12.5 micrograms glycopyrronium bromide).

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Pressurised inhalation, solution (pressurised inhalation).

Colourless to yellowish liquid solution.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

COPD

Maintenance treatment in adult patients with moderate to severe chronic obstructive pulmonary disease (COPD) who are not adequately treated by a combination of an inhaled corticosteroid and a long-acting beta2-agonist (for effects on symptoms control and prevention of exacerbations see section 5.1).

Asthma

Maintenance treatment of asthma, in adults not adequately controlled with a maintenance combination of a long-acting beta2-agonist and medium dose of inhaled corticosteroid, and who experienced one or more asthma exacerbations in the previous year.

4.2 Posology and method of administration

Posology

Adults

The recommended dose is two inhalations of Trimbow twice daily.

The maximum dose is two inhalations of Trimbow twice daily.

Patients should be advised to take Trimbow every day even when asymptomatic.

If symptoms arise in the period between doses, an inhaled, short-acting beta2-agonist should be used for immediate relief.

Asthma

When choosing the starting dose strength of Trimbow (87/5/9 micrograms), the patients' disease severity, their previous asthma therapy including the inhaled corticosteroid (ICS) dose as well as the patients' current control of asthma symptoms and risk of future exacerbation should be considered.

Special populations

Elderly

No dosage adjustment is required in elderly patients (65 years of age and older).

Renal impairment

Trimbow can be used at the recommended dose in patients with mild to moderate renal impairment. Use of Trimbow in patients with severe renal impairment or end-stage renal disease requiring dialysis, especially if associated with significant body weight reduction, should be considered only if the expected benefit outweighs the potential risk (see section 4.4 and section 5.2).

Hepatic impairment

There are no relevant data on the use of Trimbow in patients with severe hepatic impairment and the medicinal product should be used with caution in these patients (see section 4.4 and section 5.2).

Paediatric population

COPD

There is no relevant use of Trimbow in the paediatric population (under 18 years of age) for the indication of COPD.

Asthma

The safety and efficacy of Trimbow in the paediatric population (under 18 years of age) have not yet been established. No data are available.

Method of administration

For inhalation use.

To ensure proper administration of the medicinal product, the patient should be shown how to use the inhaler correctly by a physician or other healthcare professional, who should also regularly check the adequacy of the patient's inhalation technique. The patient should be advised to read the Package Leaflet carefully and follow the instructions for use as given in the leaflet.

After inhaling, patients should rinse their mouth or gargle with water without swallowing it or brush their teeth (see section 4.4 and 6.6).

Trimbow is provided with a dose counter or dose indicator on the back of the inhaler, which shows how many actuations are left. For the 60 and 120 actuation pressurised containers each time the patient presses the container a puff of the solution is released and the counter counts down by one.

For the 180 actuation pressurised container, each time the patient presses the pressurised container a puff of the solution is released and the counter rotates by a small amount; the number of puffs remaining is displayed in intervals of 20.

Patients should be advised not to drop the inhaler as this may cause the counter to count down.

For instructions for use, see section 6.6.

4.3 Contraindications

Hypersensitivity to the active substances or to any of the excipients listed in section 6.1.

4.4 Special warnings and precautions for use

Not for acute use

Trimbow is not indicated for the treatment of acute episodes of bronchospasm, or to treat an acute COPD exacerbation (i.e. as a rescue therapy).

Hypersensitivity

Immediate hypersensitivity reactions have been reported after administration of Trimbow. If signs suggesting allergic reactions occur, in particular, angioedema (including difficulties in breathing or swallowing, swelling of the tongue, lips and face), urticaria or skin rash, Trimbow should be discontinued immediately and alternative therapy instituted.

Paradoxical bronchospasm

Paradoxical bronchospasm may occur with an immediate increase in wheezing and shortness of breath after dosing. This should be treated immediately with a fast-acting inhaled bronchodilator (reliever). Trimbow should be discontinued immediately, the patient assessed and alternative therapy instituted if necessary.

Deterioration of disease

It is recommended that treatment with Trimbow should not be stopped abruptly. If patients find the treatment ineffective, they should continue treatment but medical attention must be sought. Increasing use of reliever bronchodilators indicates a worsening of the underlying condition and warrants a reassessment of the therapy. Sudden and progressive deterioration in the symptoms of COPD is potentially life-threatening and the patient should undergo urgent medical assessment.

Cardiovascular effects

Trimbow should be used with caution in patients with cardiac arrhythmias, especially third degree atrioventricular block and tachyarrhythmias (accelerated and/or irregular heart beat), idiopathic subvalvular aortic stenosis, hypertrophic obstructive cardiomyopathy, severe heart disease (particularly acute myocardial infarction, ischaemic heart disease, congestive heart failure), occlusive vascular diseases (particularly arteriosclerosis), arterial hypertension and aneurysm.

Caution should also be exercised when treating patients with known or suspected prolongation of the QTc interval (QTc > 450 milliseconds for males, or > 470 milliseconds for females), either congenital or induced by medicinal products as these patients were excluded from clinical trials with Trimbow.

Limited data in asthmatic patients with cardiovascular co-morbidities or risk-factors suggest that these patients are also at higher risk of adverse reactions like local fungal infections or dysphonia (see section 4.8). If anaesthesia with halogenated anaesthetics is planned, it should be ensured that Trimbow is not administered for at least 12 hours before the start of anaesthesia as there is a risk of cardiac arrhythmias.

Caution is also required when Trimbow is used by patients with thyrotoxicosis, diabetes mellitus, pheochromocytoma and untreated hypokalaemia.

Pneumonia in patients with COPD

An increase in the incidence of pneumonia, including pneumonia requiring hospitalisation, has been observed in patients with COPD receiving inhaled corticosteroids. There is some evidence of an increased risk of pneumonia with increasing steroid dose but this has not been demonstrated conclusively across all studies.

There is no conclusive clinical evidence for intra-class differences in the magnitude of the pneumonia risk among inhaled corticosteroid products.

Physicians should remain vigilant for the possible development of pneumonia in patients with COPD as the clinical features of such infections overlap with the symptoms of COPD exacerbations.

Risk factors for pneumonia in patients with COPD include current smoking, older age, low body mass index (BMI) and severe COPD.

Systemic corticosteroid effects

Systemic effects may occur with any inhaled corticosteroid, particularly at high doses prescribed for long periods. The daily dose of Trimbow corresponds to a medium dose of inhaled corticosteroid; furthermore, these effects are much less likely to occur than with oral corticosteroids. Possible systemic effects include: Cushing's syndrome, Cushingoid features, adrenal suppression, growth retardation, decrease in bone mineral density, cataract, glaucoma and, more rarely, a range of psychological or behavioural effects including psychomotor hyperactivity, sleep disorders, anxiety, depression or aggression (particularly in children).

Therefore, it is important that the patient is reviewed regularly and the dose of inhaled corticosteroid is reduced to the lowest dose at which effective control of asthma is maintained (see section 4.2).

Trimbow should be administered with caution in patients with active or quiescent pulmonary tuberculosis, fungal and viral infections in the airways.

Hypokalaemia

Potentially serious hypokalaemia may result from beta2-agonist therapy. This has the potential to produce adverse cardiovascular effects. Particular caution is advised in severe COPD as this effect may be potentiated by hypoxia. Hypokalaemia may also be potentiated by concomitant treatment with other medicinal products which can induce hypokalaemia, such as xanthine derivatives, steroids and diuretics (see section 4.5).

Caution is also recommended when a number of reliever bronchodilators are used. It is recommended that serum potassium levels are monitored in such situations.

Hyperglycaemia

The inhalation of formoterol may cause a rise in blood glucose levels. Therefore blood glucose should be monitored during treatment following established guidelines in patients with diabetes.

Anticholinergic effect

Glycopyrronium should be used with caution in patients with narrow-angle glaucoma, prostatic hyperplasia or urinary retention. Patients should be informed about the signs and symptoms of acute narrow-angle glaucoma and should be informed to stop using Trimbow and to contact their doctor immediately should any of these signs or symptoms develop.

Additionally, due to the anticholinergic effect of glycopyrronium, the long-term co-administration of Trimbow with other anticholinergic-containing medicinal products is not recommended (see section 4.5).

Patients with severe renal impairment

In patients with severe renal impairment, including those with end-stage renal disease requiring dialysis, especially if associated with a significant body weight reduction, Trimbow should be used only if the expected benefit outweighs the potential risk (see section 5.2). These patients should be monitored for potential adverse reactions.

Patients with severe hepatic impairment

In patients with severe hepatic impairment, Trimbow should be used only if the expected benefit outweighs the potential risk (see section 5.2). These patients should be monitored for potential adverse reactions.

Use with a spacer

Single dose pharmacokinetic data (see section 5.2) have demonstrated that in comparison to routine use without a spacer device, the use of Trimbow with the AeroChamber Plus spacer device increased the total systemic exposure (AUC_{0-t}) to glycopyrronium. However, available safety data from long-term clinical studies have not raised any significant safety concerns (see section 5.1).

Prevention of oropharyngeal infections

In order to reduce the risk of oropharyngeal candida infection, patients should be advised to rinse their mouth or gargle with water without swallowing it or brush their teeth after inhaling the prescribed dose.

Visual disturbance

Visual disturbance may be reported with systemic and topical corticosteroid use. If a patient presents with symptoms such as blurred vision or other visual disturbances, the patient should be considered for referral to an ophthalmologist for evaluation of possible causes which may include cataract, glaucoma or rare diseases such as central serous chorioretinopathy (CSCR) which have been reported after use of systemic and topical corticosteroids.

4.5 Interaction with other medicinal products and other forms of interaction

Pharmacokinetic interactions

Since glycopyrronium is eliminated mainly by the renal route, drug interaction could potentially occur with medicinal products affecting renal excretion mechanisms (see section 5.2). The effect of organic cation transport inhibition (using cimetidine as a probe inhibitor of OCT2 and MATE1 transporters) in the kidneys on inhaled glycopyrronium disposition showed a limited increase in its total systemic exposure (AUC_{0-t}) by 16% and a slight decrease in renal clearance by 20% due to co administration of cimetidine.

Beclometasone is less dependent on CYP3A metabolism than some other corticosteroids, and in general interactions are unlikely; however, the possibility of systemic effects with concomitant use of strong CYP3A inhibitors (e.g. ritonavir, cobicistat) cannot be excluded, and therefore caution and appropriate monitoring is advised with the use of such medicinal products.

Pharmacodynamic interactions

Related to formoterol

Non-cardioselective beta-blockers (including eye drops) should be avoided in patients taking inhaled formoterol. If they are administered for compelling reasons, the effect of formoterol will be reduced or abolished.

Concomitant use of other beta-adrenergic medicinal products can have potentially additive effects; therefore caution is required when other beta-adrenergic medicinal products are prescribed concomitantly with formoterol.

Concomitant treatment with quinidine, disopyramide, procainamide, antihistamines, monoamine oxidase inhibitors, tricyclic antidepressants and phenothiazines can prolong the QT interval and increase the risk of ventricular arrhythmias. In addition, L-dopa, L-thyroxine, oxytocin and alcohol can impair cardiac tolerance towards beta2-sympathomimetics.

Concomitant treatment with monoamine oxidase inhibitors, including medicinal products with similar properties such as furazolidone and procarbazine, may precipitate hypertensive reactions.

There is an elevated risk of arrhythmias in patients receiving concomitant anaesthesia with halogenated hydrocarbons.

Concomitant treatment with xanthine derivatives, steroids, or diuretics may potentiate a possible hypokalaemic effect of beta2-agonists (see section 4.4). Hypokalaemia may increase the disposition towards arrhythmias in patients who are treated with digitalis glycosides.

Related to glycopyrronium

The long-term co-administration of Trimbow with other anticholinergic-containing medicinal products has not been studied and is therefore not recommended (see section 4.4).

Excipients

Trimbow contains a small amount of ethanol. There is a theoretical potential for interaction in particularly sensitive patients taking disulfiram or metronidazole.

4.6 Fertility, pregnancy and lactation

There is no experience with or evidence of safety issues on the use of the propellant norflurane (HFA134a) during human pregnancy or lactation. However, studies on the effect of HFA134a on the reproductive function and embryofetal development in animals revealed no clinically relevant adverse effects.

Pregnancy

There are no or limited amount of data from the use of Trimbow in pregnant women.

Studies in animals have shown reproductive toxicity (see section 5.3). Glucocorticoid agents are known to cause effects in the early gestation phase, while beta2-sympathomimetic agents like formoterol have tocolytic effects. Therefore, as a precautionary measure, it is preferable to avoid the use of Trimbow during pregnancy and during labour.

Trimbow should only be used during pregnancy if the expected benefit to the patient outweighs the potential risk to the foetus. Infants and neonates born to mothers receiving substantial doses of Trimbow should be observed for adrenal suppression.

Breast-feeding

There are no relevant clinical data on the use of Trimbow during breast-feeding in humans.

Glucocorticoids are excreted in human milk. It is reasonable to assume that beclometasone dipropionate and its metabolites are also excreted into breast-milk.

It is unknown whether formoterol or glycopyrronium (including their metabolites) pass into human breast-milk but they have been detected in the milk of lactating animals. Anticholinergic agents like glycopyrronium could suppress lactation.

A decision must be made whether to discontinue breast-feeding or to discontinue/abstain from Trimbow therapy taking into account the benefit of breast-feeding for the child and the benefit of therapy for the mothers.

Fertility

No specific studies have been performed with Trimbow with regard to the safety in human fertility. Animal studies have shown impairment of fertility (see section 5.3).

4.7 Effects on ability to drive and use machines

Trimbow has no or negligible influence on the ability to drive and use machines.

4.8 Undesirable effects

Summary of the safety profile

The most frequently reported adverse reactions in patients with COPD or asthma are respectively dysphonia (0.3% and 1.5%) and oral candidiasis (0.8% and 0.3%), which are normally associated with inhaled corticosteroids; muscle spasms (0.4% and 0.2%), which can be attributed to the long-acting beta2-agonist component and dry mouth (0.4% and 0.5%), which is a typical anticholinergic effect. In asthmatic patients, adverse reactions tend to cluster during the first 3 months following initiation of therapy and become less frequent with longer-term use (after 6 months of treatment).

Tabulated list of adverse reactions

Adverse reactions associated to beclometasone dipropionate/formoterol fumarate dihydrate/glycopyrronium occurred during clinical studies and post-marketing experience as well as adverse reactions listed for the marketed individual components are provided below, listed by system organ class and frequency.

Frequencies are defined as: very common ($\geq 1/10$); common ($\geq 1/100$ to $< 1/10$); uncommon ($\geq 1/1,000$ to $< 1/100$); rare ($\geq 1/10,000$ to $< 1/1,000$); very rare ($< 1/10,000$) and not known (cannot be estimated from available data).

MedDRA system organ class	Adverse reaction	Frequency
Infections and Infestations	Pneumonia (in COPD patients) ¹ , pharyngitis ¹ , oral candidiasis, urinary tract infection ¹ , nasopharyngitis ¹	Common
	Influenza ¹ , oral fungal infection, oropharyngeal candidiasis, oesophageal candidiasis ¹ , fungal (oro)pharyngitis, sinusitis ¹ , rhinitis ¹ , gastroenteritis ¹ , vulvovaginal candidiasis ¹	Uncommon
	Lower respiratory tract infection (fungal)	Rare
Blood and lymphatic system disorders	Granulocytopenia ¹	Uncommon
	Thrombocytopenia ¹	Very rare
Immune system disorders	Dermatitis allergic ¹	Uncommon
	Hypersensitivity reactions, including erythema, lips, face, eye and pharyngeal oedema	Rare
Endocrine disorders	Adrenal suppression ¹	Very rare
Metabolism and nutrition disorders	Hypokalaemia ¹ , hyperglycaemia ¹	Uncommon
	Decreased appetite	Rare
Psychiatric disorders	Restlessness ¹	Uncommon
	Psychomotor hyperactivity ¹ , sleep disorders ¹ , anxiety ¹ , depression ¹ , aggression ¹ , behavioural changes (predominantly in children) ¹	Frequency not known
	Insomnia	Rare
Nervous system disorders	Headache	Common
	Tremor ¹ , dizziness ¹ , dysgeusia ¹ , hypoaesthesia ¹	Uncommon
	Hypersomnia	Rare
Eye disorders	Vision blurred ¹ (see also section 4.4)	Frequency not known
	Glaucoma ¹ , cataract ¹	Very rare
Ear and labyrinth disorders	Otosalpingitis ¹	Uncommon
Cardiac disorders	Atrial fibrillation, electrocardiogram QT prolonged, tachycardia, tachyarrhythmia ¹ , palpitations	Uncommon
	Angina pectoris (stable ¹ and unstable), extrasystoles (ventricular ¹ and supraventricular), nodal rhythm, sinus bradycardia	Rare
Vascular disorders	Hyperaemia ¹ , flushing ¹ , hypertension	Uncommon
	Extravasation blood,	Rare
Respiratory, thoracic and mediastinal disorders	Dysphonia	Common
	Asthmatic crisis ¹ , cough, productive cough ¹ , throat irritation, epistaxis ¹ , pharyngeal erythema	Uncommon
	Bronchospasm paradoxical ¹ , exacerbation of asthma, oropharyngeal pain, pharyngeal inflammation, dry throat	Rare
	Dyspnoea ¹	Very rare
Gastrointestinal disorders	Diarrhoea ¹ , dry mouth, dysphagia ¹ , nausea ¹ , dyspepsia ¹ , burning sensation of the lips ¹ , dental caries ¹ , (aphthous) stomatitis	Uncommon
Skin and subcutaneous tissue disorders	Rash ¹ , urticaria ¹ , pruritus ¹ , hyperhidrosis ¹	Uncommon
	Angioedema ¹	Rare
Musculoskeletal and connective tissue disorders	Muscle spasms, myalgia, pain in extremity ¹ , musculoskeletal chest pain ¹	Uncommon
	Growth retardation ¹	Very rare
Renal and urinary disorders	Dysuria ¹ , urinary retention ¹ , nephritis ¹	Rare
		Rare
General disorders and administration site	Fatigue ¹ ,	Uncommon
	Asthenia ¹	Rare

MedDRA system organ class	Adverse reaction	Frequency
conditions	Oedema peripheral ¹	Very rare
Investigations	C-reactive protein increased ¹ , platelet count increased ¹ , free fatty acids increased ¹ , blood insulin increased ¹ , blood ketone body increased ¹ , blood cortisol decreased ¹	Uncommon
	Blood pressure increased ¹ , blood pressure decreased ¹	Rare
	Bone density decreased ¹	Very rare

¹ Adverse reactions reported in the SmPC of at least one of the individual components, but not observed as adverse reactions in the clinical development of Trimbow

Among the observed adverse reactions the following are typically associated with:

Beclometasone dipropionate: pneumonia, oral fungal infections, lower respiratory tract infection fungal, dysphonia, throat irritation, hyperglycaemia, psychiatric disorders, blood cortisol decreased, blurred vision.

Formoterol: hypokalaemia, hyperglycaemia, tremor, palpitations, muscle spasms, electrocardiogram QT prolonged, blood pressure increased, blood pressure decreased, atrial fibrillation, tachycardia, tachyarrhythmia, angina pectoris (stable and unstable), ventricular extrasystoles, nodal rhythm.

Glycopyrronium: glaucoma, atrial fibrillation, tachycardia, palpitations, dry mouth, dental caries, dysuria, urinary retention, urinary tract infection.

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via the national reporting system.

4.9 Overdose

An overdose of Trimbow may produce signs and symptoms due to the individual component's actions, including those seen with overdose of other beta2-agonists or anticholinergics and consistent with the known inhaled corticosteroid class effects (see section 4.4). If overdose occurs, the patient's symptoms should be treated supportively with appropriate monitoring as necessary.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotheapeutic group: Drugs for obstructive airway diseases, adrenergics in combinations with anticholinergics. ATC code: R03AL09.

Mechanism of action and pharmacodynamic effects

Trimbow contains beclometasone dipropionate, formoterol and glycopyrronium in a solution formulation resulting in an aerosol with extrafine particles with an average mass median aerodynamic diameter (MMAD) of around 1.1 micrometres and co-deposition of the three components. The aerosol particles of Trimbow are on average much smaller than the particles delivered in non-extrafine formulations. For beclometasone dipropionate, this results in a more potent effect than formulations with a non-extrafine particle size distribution (100 micrograms of beclometasone dipropionate extrafine in Trimbow are equivalent to 250 micrograms of beclometasone dipropionate in a non-extrafine formulation).

Beclometasone dipropionate

Beclometasone dipropionate given by inhalation at recommended doses has a glucocorticoid anti-inflammatory action within the lungs. Glucocorticoids are widely used for the suppression of inflammation in chronic inflammatory diseases of the airways. Their action is mediated by the binding to glucocorticoid

receptors in the cytoplasm resulting in the increased transcription of genes coding for antiinflammatory proteins.

Formoterol

Formoterol is a selective beta2-adrenergic agonist that produces relaxation of bronchial smooth muscle in patients with reversible airways obstruction. The bronchodilating effect sets in rapidly, within 1-3 minutes after inhalation, and has a duration of 12 hours after a single dose.

Glycopyrronium

Glycopyrronium is a high-affinity, long-acting muscarinic receptor antagonist (anticholinergic) used for inhalation as bronchodilator treatment. Glycopyrronium works by blocking the bronchoconstrictor action of acetylcholine on airway smooth muscle cells, thereby dilating the airways. Glycopyrronium bromide is a high affinity muscarinic receptor antagonist with a greater than 4-fold selectivity for the human M3 receptors over the human M2 receptor as it has been demonstrated.

Clinical efficacy and safety

COPD

The Phase III clinical development programme in COPD included two 52-week active-controlled studies. Study TRILOGY compared Trimbow with a fixed combination of beclometasone dipropionate and formoterol 100/6 micrograms two inhalations twice daily (1,368 randomised patients). Study TRINITY compared Trimbow with tiotropium 18 micrograms inhalation powder, hard capsule, one inhalation once daily; in addition, effects were compared with an extemporaneous triple combination made of a fixed combination of beclometasone dipropionate and formoterol 100/6 micrograms two inhalations twice daily plus tiotropium 18 micrograms inhalation powder, hard capsule, one inhalation once daily (2,691 randomised patients). Both studies were conducted in patients with a clinical diagnosis of COPD with severe to very severe airflow limitation (FEV₁ less than 50% predicted), with symptoms assessed as a COPD Assessment Test (CAT) score of 10 or above, and with at least one COPD exacerbation in the previous year. The two studies included approximately 20% of patients who used the AeroChamber Plus spacer.

Reduction of COPD exacerbations

Compared with a fixed combination of beclometasone dipropionate and formoterol, Trimbow reduced the rate of moderate/severe exacerbations over 52 weeks by 23% (rate: 0.41 versus 0.53 events per patient/year; $p = 0.005$). Compared with tiotropium, Trimbow reduced the rate of moderate/severe exacerbations over 52 weeks by 20% (rate: 0.46 versus 0.57 events per patient/year; $p = 0.003$). Compared with tiotropium, Trimbow also reduced the rate of severe exacerbations (i.e. excluding moderate exacerbations) by 32% (rate: 0.067 versus 0.098 events per patient/year; $p = 0.017$). No differences were observed when comparing Trimbow and the extemporaneous triple combination (moderate/severe exacerbation rate: 0.46 versus 0.45 events per patient/year).

In addition, compared with both a fixed combination of beclometasone dipropionate and formoterol and with tiotropium, Trimbow significantly prolonged the time to first exacerbation (hazard ratio 0.80 and 0.84 respectively; $p = 0.020$ and 0.015 respectively), with no differences between Trimbow and the extemporaneous triple combination (hazard ratio 1.06).

Effects on lung function

Pre-dose FEV₁

Compared with a fixed combination of beclometasone dipropionate and formoterol, Trimbow improved pre-dose FEV₁ by 81 mL after 26 weeks of treatment and by 63 mL after 52 weeks of treatment. Compared with tiotropium, Trimbow improved pre-dose FEV₁ by 51 mL after 26 weeks of treatment and by 61 mL after 52 weeks of treatment. These improvements were statistically significant ($p < 0.001$).

No differences were observed when comparing Trimbow and the extemporaneous triple combination (difference of 3 mL in pre-dose FEV₁ after 52 weeks of treatment).

2-hour post-dose FEV₁

Compared with a fixed combination of beclometasone dipropionate and formoterol, Trimbow significantly improved 2-hour post dose FEV₁ by 117 mL after 26 weeks of treatment and by 103 mL after 52 weeks of treatment ($p < 0.001$). This endpoint was only measured in study TRILOGY.

Inspiratory Capacity (IC)

Compared with tiotropium, Trimbow significantly improved IC by 39 mL ($p = 0.025$) and 60 mL ($p = 0.001$) after 26 and 52 weeks of treatment respectively. Similar effects were seen when comparing Trimbow with the extemporaneous triple combination. This endpoint was only measured in study TRINITY.

Symptomatic outcomes

Trimbow significantly improved dyspnoea (measured as the Transition Dyspnoea Index – TDI - focal score) after 26 weeks of treatment compared with baseline (by 1.71 units; $p < 0.001$), but the adjusted mean difference versus a fixed combination of beclometasone dipropionate and formoterol was not statistically significant (0.21 units; $p = 0.160$). A responder analysis showed that a significantly greater percentage of patients had a clinically significant improvement (focal score greater than or equal to 1) after 26 weeks with Trimbow than with a fixed combination of beclometasone dipropionate and formoterol (57.4% versus 51.8%; $p = 0.027$). TDI was only measured in study TRILOGY.

Trimbow was also statistically significantly superior to both a fixed combination of beclometasone dipropionate and formoterol and to tiotropium in terms of improvement in quality of life (measured by the Saint George Respiratory Questionnaire – SGRQ - total score). A responder analysis showed that a significantly greater percentage of patients had a clinically significant improvement (reduction versus baseline of greater than or equal to 4) after 26 and 52 weeks with Trimbow than with a fixed combination of beclometasone dipropionate and formoterol and with tiotropium.

Asthma

The Phase III clinical development programme in asthma included a randomized, double-blind, active-controlled study of 52 weeks duration (TRIMARAN), performed with the medium ICS dose strength (Trimbow 87/5/9)

The TRIMARAN study was conducted in adult patients with a clinical diagnosis of asthma who were uncontrolled on dual maintenance treatment using medium dose ICS/LABA combination (ACQ-7 score ≥ 1.5). In order to be eligible, patients had to have experienced at least one asthma exacerbation requiring treatment with systemic corticosteroids or emergency department visit or in-patient hospitalisation in the previous year.

The TRIMARAN study compared two twice-daily doses of BDP/FF/G 87/5/9 (N=579) with two twice-daily doses of fixed dose combination of Beclometasone dipropionate (BDP) and formoterol (FF) 100/6 micrograms (delivered dose of 84.6/5.0 (N=576).

The primary objective of the study was to demonstrate superiority of BDP/FF/G 87/5/9 (two inhalations twice daily) over the respective fixed dual combination product (medium dose ICS/LABA) in terms of the co-primary endpoints (change from baseline in pre-dose FEV₁ at Week 26 and the rate of moderate and severe exacerbation rate over 52 weeks).

Median age of patients enrolled in the TRIMARAN pivotal study was 54 years. Less than 20% of patients were 65 years or older and approximately 60% of patients were female. During the study, about 16% of patients used the AeroChamber Plus spacer.

Reduction of asthma exacerbations

In the TRIMARAN study, Trimbow 87/5/9 significantly reduced the rate of moderate/severe exacerbations compared with BDP/FF 84.6/5.0 (adjusted rate ratio 0.846, 95%CI [0.725; 0.987]) (Table 1).

Additionally, the time to first moderate/severe exacerbation (secondary endpoint) was prolonged in the triple combination arm when compared with the respective dual combination arm.

Effects on lung function

In the TRIMARAN study, Trimbow 87/5/9 improved the lung function parameters of pre-dose FEV₁ (co-primary endpoint), peak_{0-3h} FEV₁, and morning peak expiratory flow (key secondary endpoints), compared with a fixed combination of beclometasone dipropionate and formoterol 100/6 micrograms and 200/6 micrograms, respectively, after 26 weeks of treatment. All improvements were statistically significant (see Table 1).

Table 1 - Results of primary and secondary endpoints (ITT population)

Study	TRIMARAN
Comparison of interest	Trimbow 87/5/9 (N=579)
N = randomised patients per treatment arm	vs BDP/FF¹ 84.6/5 (N=576)
Primary endpoints	
<i>Pre-dose FEV₁ after 26 weeks (co-primary endpoint)</i>	
Treatment difference	+57 mL
p-value	p = 0.008
<i>Moderate/severe exacerbations over 52 weeks (co-primary endpoint)</i>	
Adjusted rate per patient/year	1.83 vs 2.16
Rate change	-15.4%
p-value	p = 0.033
Key secondary and secondary endpoints	
<i>Peak_{0-3h} FEV₁ after 26 weeks (key secondary endpoint)</i>	
Treatment difference	+84 mL
p-value	p < 0.001
<i>Morning peak expiratory flow (PEF) over 26 weeks (key secondary endpoint)</i>	
Treatment difference	+8 L/min
p-value	p < 0.001
<i>Time to the first moderate/severe exacerbation over 52 weeks (secondary endpoint)</i>	
Hazard ratio	0.84
p-value	p = 0.022*

Co-primary endpoints (pre-dose FEV₁ at Week 26 and the rate of moderate and severe exacerbation rate over 52 weeks) and the key secondary endpoints (peak_{0-3h} FEV₁ at Week 26, morning PEF over 26 weeks) were part of the hierarchical confirmatory testing strategy and thus were controlled for multiplicity.

¹ = fixed combination of beclometasone dipropionate (BDP) plus formoterol fumarate (FF)

* = nominal p-values (not included in the hierarchical testing strategy)

Paediatric population

The European Medicines Agency has waived the obligation to submit the results of studies with Trimbow in all subsets of the paediatric population in COPD (see section 4.2 for information on paediatric use).

The safety and efficacy of Trimbow in children and adolescents with asthma under 18 years of age have not yet been established (see section 4.2 for information on paediatric use).

5.2 Pharmacokinetic properties

Related to Trimbow

The systemic exposure to beclometasone dipropionate, formoterol and glycopyrronium has been investigated in a pharmacokinetic study conducted in healthy subjects. The study compared data obtained after treatment with a single dose of Trimbow (4 inhalations of 100/6/25 micrograms, a non-marketed formulation containing twice the approved strength of glycopyrronium) or a single dose of the extemporaneous combination of beclometasone dipropionate/formoterol (4 inhalations of 100/6 micrograms) plus glycopyrronium (4 inhalations of 25 micrograms). The maximum plasma concentration and systemic exposure of beclometasone dipropionate main active metabolite (beclometasone 17-monopropionate) and formoterol were similar after administration of the fixed or extemporaneous combination. For glycopyrronium, the maximum plasma concentration was similar after administration of the fixed or extemporaneous combination, while the systemic exposure was slightly higher after administration of Trimbow than with the extemporaneous combination. This study also investigated the potential pharmacokinetic interaction between the active components of Trimbow by comparing the pharmacokinetic data obtained after a single dose of the extemporaneous combination or after a single dose of the single components beclometasone dipropionate/formoterol or glycopyrronium. There was no clear evidence of pharmacokinetic interaction,

however the extemporaneous combination showed formoterol and glycopyrronium levels transiently slightly higher immediately after dosing compared with the single components. It is noted that single component glycopyrronium, formulated as pressurised metered dose inhaler, which was used in the PK studies, is not available on the market.

A comparison across studies showed that the pharmacokinetics of beclometasone 17-monopropionate, formoterol and glycopyrronium is similar in COPD patients and in healthy subjects.

Effect of a spacer

The use of Trimbow with the AeroChamber Plus spacer in COPD patients increased the lung delivery of beclometasone 17-monopropionate, formoterol and glycopyrronium (maximum plasma concentration increased by 15%, 58% and 60% respectively). The total systemic exposure (as measured by AUC_{0-t}) was slightly reduced for beclometasone 17-monopropionate (by 37%) and formoterol (by 24%), while it was increased for glycopyrronium (by 45%). See also section 4.4.

Effect of renal impairment

Systemic exposure (AUC_{0-t}) to beclometasone dipropionate, to its metabolite beclometasone 17-monopropionate and to formoterol was not affected by mild to severe renal impairment. For glycopyrronium, there was no impact in subjects with mild and moderate renal impairment. However, an increase in total systemic exposure of up to 2.5-fold was observed in subjects with severe renal impairment (glomerular filtration rate below 30 mL/min/1.73 m²), as a consequence of a significant reduction of the amount excreted in urine (approximately 90% reduction of glycopyrronium renal clearance). Simulations performed with a pharmacokinetic model showed that even when covariates had extreme values (body weight less than 40 kg and concomitant glomerular filtration rate below 27 mL/min/1.73 m²), exposure to Trimbow active substances remains in approximately a 2.5-fold range compared to the exposure in a typical patient with median covariate values.

Related to beclometasone dipropionate

Beclometasone dipropionate is a pro-drug with weak glucocorticoid receptor binding affinity that is hydrolysed via esterase enzymes to an active metabolite beclometasone 17-monopropionate which has a more potent topical anti-inflammatory activity compared with the pro-drug beclometasone dipropionate.

Absorption, distribution and biotransformation

Inhaled beclometasone dipropionate is rapidly absorbed through the lungs; prior to absorption there is extensive conversion to beclometasone 17-monopropionate via esterase enzymes that are found in most tissues. The systemic availability of the active metabolite arises from lung (36%) and from gastrointestinal absorption of the swallowed dose. The bioavailability of swallowed beclometasone dipropionate is negligible; however, pre-systemic conversion to beclometasone 17-monopropionate results in 41% of the dose being absorbed as the active metabolite. There is an approximately linear increase in systemic exposure with increasing inhaled dose. The absolute bioavailability following inhalation is approximately 2% and 62% of the nominal dose for unchanged beclometasone dipropionate and beclometasone 17-monopropionate respectively. Following intravenous dosing, the disposition of beclometasone dipropionate and its active metabolite is characterised by high plasma clearance (150 and 120 L/h respectively), with a small volume of distribution at steady state for beclometasone dipropionate (20 L) and larger tissue distribution for its active metabolite (424 L). Plasma protein binding is moderately high.

Elimination

Faecal excretion is the major route of beclometasone dipropionate elimination mainly as polar metabolites. The renal excretion of beclometasone dipropionate and its metabolites is negligible. The terminal elimination half-lives are 0.5 hours and 2.7 hours for beclometasone dipropionate and beclometasone 17-monopropionate respectively.

Patients with hepatic impairment

The pharmacokinetics of beclometasone dipropionate in patients with hepatic impairment has not been studied, however, as beclometasone dipropionate undergoes a very rapid metabolism via esterase enzymes present in intestinal fluid, serum, lungs and liver to form the more polar products beclometasone

21-monopropionate, beclometasone 17-monopropionate and beclometasone, hepatic impairment is not expected to modify the pharmacokinetics and safety profile of beclometasone dipropionate.

Related to formoterol

Absorption and distribution

Following inhalation, formoterol is absorbed from both the lung and the gastrointestinal tract. The fraction of an inhaled dose that is swallowed after administration with a metered dose inhaler may range between 60% and 90%. At least 65% of the fraction that is swallowed is absorbed from the gastrointestinal tract. Peak plasma concentrations of the unchanged active substance occur within 0.5 to 1 hours after oral administration. Plasma protein binding of formoterol is 61-64% with 34% bound to albumin. There was no saturation of binding in the concentration range attained with therapeutic doses. The elimination half-life determined after oral administration is 2-3 hours. Absorption of formoterol is linear following inhalation of 12 to 96 micrograms of formoterol.

Biotransformation

Formoterol is widely metabolised and the prominent pathway involves direct conjugation at the phenolic hydroxyl group. Glucuronide acid conjugate is inactive. The second major pathway involves O-demethylation followed by conjugation at the phenolic 2'-hydroxyl group. Cytochrome P450 isoenzymes CYP2D6, CYP2C19 and CYP2C9 are involved in the O-demethylation of formoterol. Liver appears to be the primary site of metabolism. Formoterol does not inhibit CYP450 enzymes at therapeutically relevant concentrations.

Elimination

The cumulative urinary excretion of formoterol after single inhalation from a dry powder inhaler increased linearly in the 12-96 micrograms dose range. On average, 8% and 25% of the dose was excreted as unchanged and total formoterol, respectively. Based on plasma concentrations measured following inhalation of a single 120 micrograms dose by 12 healthy subjects, the mean terminal elimination half-life was determined to be 10 hours. The (R,R)- and (S,S)-enantiomers represented about 40% and 60% of unchanged active substance excreted in the urine, respectively. The relative proportion of the two enantiomers remained constant over the dose range studied and there was no evidence of relative accumulation of one enantiomer over the other after repeated dosing. After oral administration (40 to 80 micrograms), 6% to 10% of the dose was recovered in urine as unchanged active substance in healthy subjects; up to 8% of the dose was recovered as the glucuronide. A total 67% of an oral dose of formoterol is excreted in urine (mainly as metabolites) and the remainder in the faeces. The renal clearance of formoterol is 150 mL/min.

Patients with hepatic impairment

The pharmacokinetics of formoterol has not been studied in patients with hepatic impairment; however, as formoterol is primarily eliminated via hepatic metabolism, an increased exposure can be expected in patients with severe hepatic impairment.

Related to glycopyrronium

Absorption and distribution

Glycopyrronium has a quaternary ammonium structure which limits its passage across biological membranes and produces slow, variable and incomplete gastrointestinal absorption. Following glycopyrronium inhalation, the lung bioavailability was 10.5% (with activated charcoal ingestion) while the absolute bioavailability was 12.8% (without activated charcoal ingestion) confirming the limited gastrointestinal absorption and indicating that more than 80% of glycopyrronium systemic exposure was from lung absorption. After repeated inhalation of twice daily doses ranging from 12.5 to 50 micrograms via pressurised metered dose inhaler in COPD patients, glycopyrronium showed linear pharmacokinetics with little systemic accumulation at steady state (median accumulation ratio 2.2-2.5).

The apparent volume of distribution (V_z) of inhaled glycopyrronium was increased compared to intravenous (i.v.) infusion (6420 L versus 323 L), reflecting the slower elimination after inhalation.

Biotransformation

The metabolic pattern of glycopyrronium *in vitro* (humans, dogs, rats, mice and rabbits liver microsomes and hepatocytes) was similar among species and the main metabolic reaction was the hydroxylation on the

phenyl or cyclopentyl rings. CYP2D6 was found to be the only enzyme responsible for glycopyrronium metabolism.

Elimination

The mean elimination half-life of glycopyrronium in healthy volunteers was approximately 6 hours after i.v. injection while after inhalation in COPD patients it ranged from 5 to 12 hours at steady state. After a glycopyrronium single i.v. injection, 40% of the dose was excreted in the urine within 24 hours. In COPD patients receiving repeated twice daily administration of inhaled glycopyrronium, the fraction of the dose excreted in urine ranged from 13.0% to 14.5% at steady state. Mean renal clearance was similar across the range of doses tested and after single and repeated inhalation (range 281-396 mL/min).

5.3 Preclinical safety data

Non-clinical data reveal no special hazard for humans based on conventional studies of safety pharmacology, repeat dose toxicity and toxicity to reproduction and development.

Safety pharmacology

In an inhalation study in telemetered dogs, the cardiovascular system was a major target system for acute effects of Trimbow (increase in heart rate, decrease in blood pressure, ECG changes at higher doses), effects probably mainly related to the beta2-adrenergic activity of formoterol and the anti-muscarinic activity of glycopyrronium. There was no evidence for overadditive effects of the triple combination when compared with the single components.

Repeat dose toxicity

In repeat dose inhalation studies with Trimbow in rats and dogs of up to 13 weeks duration, the main observed alterations were related to effects on the immune system (probably due to systemic corticosteroid effects of beclometasone dipropionate and its active metabolite beclometasone-17-monopropionate) and on the cardiovascular system (probably related to the beta2-adrenergic activity of formoterol and the anti-muscarinic activity of glycopyrronium). The toxicological profile of the triple combination reflected that of the single active components without a relevant increase in toxicity and without unexpected findings.

Reproductive and development toxicity

Beclometasone dipropionate/beclometasone-17-monopropionate was considered responsible for reproductive toxicity effects in rats such as reduction of the conception rate, fertility index, early embryonic development parameters (implantation loss), delay in ossification and increased incidence of visceral variations; while tocolytic and anti-muscarinic effects, attributed to the beta2-adrenergic activity of formoterol and the anti-muscarinic activity of glycopyrronium, affected pregnant rats in the late phase of gestation and/or early phase of lactation, leading to loss of pups.

Genotoxicity

Genotoxicity of Trimbow has not been evaluated, however, the single active components were devoid of genotoxic activity in the conventional test systems.

Carcinogenicity

Carcinogenicity studies have not been performed with Trimbow. However, in a 104-week rat inhalation carcinogenicity study and an oral 26-week carcinogenicity study in transgenic Tg.rasH2 mice, glycopyrronium bromide showed no carcinogenic potential and published data concerning long-term studies conducted with beclometasone dipropionate and formoterol fumarate in rats do not indicate a clinically relevant carcinogenic potential.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Ethanol anhydrous
Hydrochloric acid
Norflurane (propellant)

6.2 Incompatibilities

Not applicable.

6.3 Shelf life

Prior to dispensing, the medicinal product may be stored 18 months in a refrigerator (2-8°C). Chemical and physical in-use stability has been demonstrated for 2 months at 30°C.

After dispensing, the medicinal product may be stored for a maximum of 2 months at a temperature up to 30°C. Other in-use storage times and conditions are under the responsibility of the user.

6.4 Special precautions for storage

Prior to dispensing:

Store in a refrigerator (2°C-8°C).

Do not freeze.

Do not expose to temperatures higher than 50°C.

Do not pierce the pressurised container.

For in-use storage conditions, see section 6.3.

6.5 Nature and contents of container

Pressurised container (coated aluminium), with a metering valve. The pressurised container is inserted in a polypropylene inhaler which incorporates a mouthpiece and a dose counter (60 actuations or 120 actuations per pressurised container) or dose indicator (180 actuations per pressurised container) and is provided with a polypropylene mouthpiece cap.

Pack sizes of 1 container of 60, 120 or 180 actuations and multipacks of 2 or 3 containers of 120 actuations each.

Not all pack sizes may be marketed.

6.6 Special precautions for disposal and other handling

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

For pharmacists:

Enter the date of dispensing to the patient on the pack.

Instructions for use

Priming the inhaler

Before using the inhaler for the first time, the patient should release one actuation into the air in order to ensure that the inhaler is working properly (priming). Before priming the 60, 120 or 180 actuation

pressurised containers, the counter/indicator should read 61, 121 or 180, respectively. After priming, the counter/indicator should read 60, 120 or 180.

Use of the inhaler

Patients should stand or sit in an upright position when inhaling from their inhaler. The steps below should be followed. IMPORTANT: steps 2 to 5 should not be performed too quickly:

1. Patients should remove the protective cap from the mouthpiece and check that the mouthpiece is clean and free from dust and dirt or any other foreign objects.
2. Patients should breathe out slowly and as deeply as comfortable, in order to empty their lungs.
3. Patients should hold the inhaler vertically with its body upwards and place the mouthpiece between their teeth without biting. Their lips should then be placed around the mouthpiece, with the tongue flat under it.
4. At the same time, patients should breathe in slowly and deeply through the mouth until the lungs are full of air (this should take approximately 4 – 5 seconds). Immediately after starting to breathe in, patients should firmly press down on the top of the pressurised container to release one puff.
5. Patients should then hold their breath for as long as comfortably possible, then remove the inhaler from the mouth and breathe out slowly. Patients should not breathe out into the inhaler.
6. Patients should then check the dose counter or dose indicator to ensure it has moved accordingly.

To inhale the second puff, patients should keep the inhaler in a vertical position for approximately 30 seconds and repeat steps 2 to 6.

If mist appears after the inhalation, either from the inhaler or from the sides of the mouth, the procedure should be repeated from step 2.

After use, patients should close the inhaler with the protective mouthpiece cover and check the dose counter or dose indicator.

After inhaling, patients should rinse their mouth or gargle with water without swallowing it or brush their teeth (see sections 4.2 and 4.4).

When to get a new inhaler

Patients should be advised to get a new inhaler when the dose counter or indicator shows the number 20. They should stop using the inhaler when the counter or indicator shows 0 as any puffs left in the device may not be enough to release a full actuation.

Additional instructions for specific groups of patients

For patients with weak hands it may be easier to hold the inhaler with both hands. Therefore, the index fingers should be placed on the top of the pressurised container and both thumbs on the base of the inhaler.

Patients who find it difficult to synchronise aerosol actuation with inspiration of breath may use the AeroChamber Plus spacer device, properly cleaned as described in the relevant leaflet. They should be advised by their doctor or pharmacist about the proper use and care of their inhaler and spacer and their technique checked to ensure optimum delivery of the inhaled active substance to the lungs. This may be obtained by the patients using the AeroChamber Plus by one continuous slow and deep breath through the spacer, without any delay between actuation and inhalation. Alternatively, patients may simply breathe in and out (through the mouth) after the actuation, as instructed in the spacer leaflet, to obtain the active substance. See section 4.4 and section 5.2.

Cleaning

For the regular cleaning of the inhaler, patients should remove weekly the cap from the mouthpiece and wipe the outside and inside of the mouthpiece with a dry cloth. They should not remove the pressurised container from the actuator and should not use water or other liquids to clean the mouthpiece.

7. MANUFACTURER

Chiesi Farmaceutici S.p.A.

Via San Leonardo 96
43122 Parma Italy

8. PRODUCT REGISTRATION HOLDER

Orient EuroPharma (M) Sdn Bhd
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Malaysia

9. DATE OF REVISION OF THE TEXT

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